Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/27/2007
PI Name:	Tittel, Frank K Ph.D.		
Project Title:	Development of Advanced Mid-Infrared Laser Based Gas Sensor Technology		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77005	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kosterev, Anatoliy Ph.D. (Rice University) Bakhirkin, Yury Ph.D. (Rice University) Wysocki, Gerard Ph.D. (Rice University)		
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	This project is a joint Rice University- George Washington University-JPL research and development program of mid-infrared diode laser sources in the 3 to 4.4 micron region and their use in trace gas sensing for NASA applications relevant to spacecraft environmental monitoring and advanced life support. At Rice the primary emphasis will be to demonstrate and emphasize a novel optical detection technique based on an ultra-compact photoacoustic analyzer design that will significantly enhance the performance and robustness of near infrared and mid-infrared gas sensors. The group of Prof. Houston Miller is providing expertise in spectroscopic strategy for detecting Code M molecular target gases as well as developing a cavity ringdown spectroscopy (CRDS) based sensor technology. JPL with semiconductor facilities and the necessary expertise is developing Sb-based diode and type II interband cascade lasers with distributed feedback structures for meeting the spectroscopic application needs.		

Task Description:	The primary emphasis of this task is the development of compact, robust trace gas sensor technology based on quartz enhanced photoacoustic spectroscopy that can be applied to the sensitive, selective and real-time detection of Code M molecular targets (formaldehyde, acetaldehyde, ammonia, carbon monoxide, hydrogen cyanide, carbon dioxide, nitrogene dioxide, HF, acrolein, and water vapor) at specified detection limits for each of the target gases. Manned space missions planned for this century will include longer stays on the International Space Station, expeditions to Mars, as well as return trips to the moon. Hence there is a need to create and maintain an environment suitable for human habitation. This will require the development of instrumentation capable of precise detection, quantifying and controlling a number of molecular trace gas species that are relevant to fire detection and the air quality of spacecraft habitats.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Compact, sensitive and selective gas sensors based on near infrared and mid infrared semiconductor lasers and quartz enhanced photoacoustic spectroscopy. This technology is applicable to atmospheric chemistry, environmental monitoring, chemical analysis and industrial prcess control as well as medical diagnostics based on breath analysis.
Task Progress:	Influence of Molecular Relaxation Dynamics on Quartz-Enhanced Photoacoustic Detection of CO2 at a wavelength of 2 microns Carbon dioxide (CO2) trace gas detection based on quartz enhanced photoacoustic spectroscopy (QEPAS) using a distributed feedback diode laser operating at 2 microns was performed, with a primary purpose of studying vibrational relaxation processes in the CO2-N2-H2O system [1]. A simple model was developed and used to explain the experimentally observed dependence of amplitude and phase of the photoacoustic signal on pressure and gas humidity. A (1sigma) sensitivity of 110 parts-per-million (with a 1s lock-in time constant) was obtained for CO2 concentration measured in humid gas samples. The strongest CO2 molecular absorption lines ~ 2 microns are in the R branch of the 2 nu1 + nu3 band. Diode laser selected for this experiment (NTT Electronics, model: KELD1G5B2TA) can be tuned between 4982 cm-1 and 4993 cm-1 by changing the temperature of the laser submount between 41oC and 12oC respectively. Fine laser frequency tuning can be performed by changing laser current with a wavelength tuning coefficient of . HITRAN 2000 simulation was used to examine CO2 absorption features within the tuning range of the laser radiation together with possible interference from H2O. The calculation was done for average atmosphere CO2 concentrations of 350 ppm and 100% humidity for an absorption pathlength of 1m, and a pressure of 300 Torr which was found to be optimum for CO2 QEPAS-based monitoring in dry air. The strongest lines R16 and R18 with line strength of and respectively, are free of H2O interference and can be used for sensitive CO2 concentration measurements. Within the overall laser tuning range one can also target several ammonia (NH3) absorption lines with a line strength of the order of (~6 times stronger then the available CO2 lines), which can be used for simultaneous multi-species measurement with a single diode laser source. 2. Detection Of Formaldehyde Using Off-Axis Integrated Cavity Output Spectroscopy With An Interband Cascade Laser A continuous-wave, mid-infrared, distributed feedback, interband cascade laser was used to detect and quantify formaldehyde (H2CO) using off axis, integrated cavity output spectroscopy in gas mixtures ranging from ~ 1-25 ppm in 106 ppmV in H2CO [2]. Analysis of the spectral measurements indicate that a H2CO concentration of 150 ppbV would produce a spectrum with a signal to noise ratio of 3 for a data acquisition time of 3 sec. This is a relevant sensitivity level for formaldehyde monitoring of indoor air, occupational settings, and on board spacecraft in long duration missions in particular as the detection sensitivity improves with the square root of the data acquisition time A committee of the National Academy of Sciences, working on behalf of NASA, has extensively reviewed the toxicity of formaldehyde and has established Spacecraft Maximum Allowable Concentrations for various times of exposure [3]. Their recommended upper limits range from 0.4 ppmV for short term (1 hr) exposure down to 0.04 ppmV for 7 to 180 day space missions. The optical arrangement and electronic control of the experiment for ICOS is similar to that published previously by the Rice Laser Science group for measurement of NO at 1912 cm-1 using a quantum cascade laser [4]. The interband cascade laser used is identical to that described in a recent publication in which formaldehyde was measured utilizing quartz-enhanced photo acoustic spectroscopy [5]. Briefly, light from a cooled ICL was collimated using a 12.5 mm f.l. ZnSe lens and directed into an ICOS cell. Downstream output from the ICOS cell was collected with an off-axis parabolic mirror (75 mm diameter; 75 mm f.l.) and detected with an InSb detector. The ICL, housed in a liquid-nitrogen cryostat provided output powers of up to 12 mW at 3.53 µm. The output is tunable from 2831.8 cm-1 to 2833.7 cm-1 by varying the injection current. The low threshold current operation of ICLs requires a current source with relatively high precision (an ILX Lightwave model LDX 3220 was used in this study). The specific ICL used in this work is able to operate continuously up to 170K. At or near liquid nitrogen temperatures, the laser operates with a bias voltage of ~7-8 V and a relatively low threshold current of approximately 3 mA. Although the output of the DFB ICL laser is nominally single mode, other weaker modes are apparent in the spectrum of the laser output at ~2840, 2851, and 2866 cm-1 whose power can be up to ~ 4%, 2%, and 1% of the power of the 2832 cm-1 mode. 3. Sub-ppbv Nitric Oxide Concentration Measurements using CW Thermoelectrically Cooled Quantum Cascade Laser-Based Integrated Cavity Output Spectroscopy A nitric oxide (NO) gas sensor based on a thermoelectrically cooled, continuous wave, distributed feedback quantum cascade laser operating at 5.45 microns (1835 cm-1), and off-axis integrated cavity output spectroscopy (OA-ICOS) combined with a wavelength modulation technique was developed to determine NO concentrations at sub-ppbv levels [6]. Such a sensor is important for a number of applications, such as medical diagnostics, environmental monitoring, and industrial process control. The sensor employs a 50 cm long high-finesse optical cavity that provides an effective pathlength of ~ 700 m. A noise equivalent (SNR = 1) minimum detection limit of 0.7 ppbv with a 1 second observation time was achieved. References:

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